Activation Functions in Neural Networks

1. What is an Activation Function?

An activation function is a mathematical function applied to the weighted sum of a neuron. It transforms the neuron's input into an output that is passed to the next layer.

A neuron first calculates its weighted sum and bias:

$$z = \Sigma (w_i x_i) + b$$

The activation function then transforms z into the neuron's output. Here, x_i is an input, w_i is its weight, b is the bias, and z is the pre-activation value.

Inputs → Weighted Sum + Bias → Activation Function → Output

2. Need for Activation Functions

Introduce non-linearity: Real-world problems are rarely purely linear. Non-linear activation functions allow neural networks to learn complex relationships.

Enable deep learning: Stacking layers with non-linear activations lets a network learn increasingly complex representations.

Create useful decision boundaries: Non-linear activations allow models to represent complex classification boundaries.

Control neuron outputs: Activation functions can transform raw neuron values into useful output ranges.

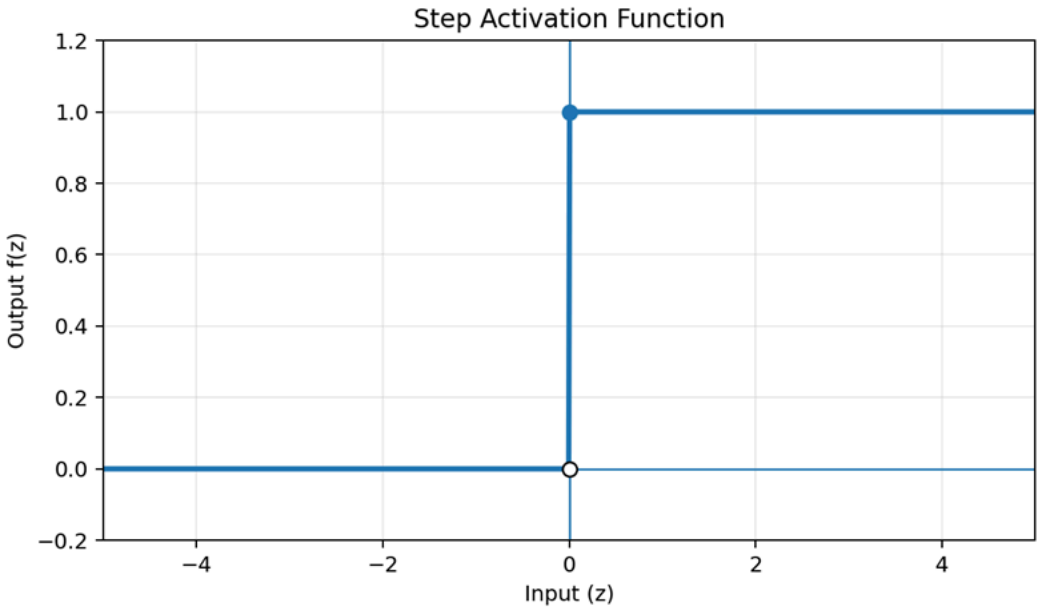
Why only linear layers are not enough

If every layer in a neural network performs only a linear transformation, the composition of those layers can still be represented as a single linear transformation. Therefore, simply adding more linear layers does not give the network the ability to learn complex non-linear patterns.

3. Step Activation Function

The Step activation function produces a discrete output based on whether the input crosses a threshold. A common binary Step function returns 0 for negative inputs and 1 for non-negative inputs.

$$f(z) = 1 \text{ if } z \geq 0 \quad ; \quad f(z) = 0 \text{ if } z < 0$$



How it works

- For a negative input, the output is 0.
- For a non-negative input, the output is 1.
- The threshold creates a hard binary decision.

Example

If a perceptron produces z = 2.5, the Step function returns 1. If it produces z = −1.2, the Step function returns 0.

Advantages

- Simple and easy to understand.
- Produces a clear binary decision.

- Historically important in the perceptron.

Limitations

- Not differentiable at the threshold.
- Its derivative is zero almost everywhere.
- It does not provide a useful gradient for standard backpropagation-based training.

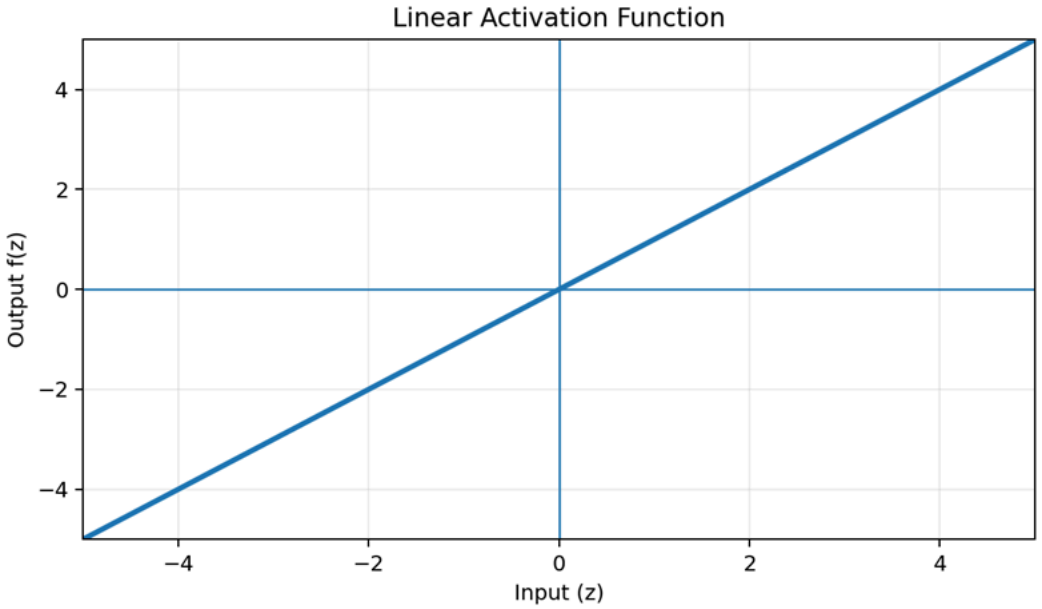
Typical Use

The Step function is mainly used for explaining the classical perceptron and hard binary decisions. It is generally not used in modern deep networks trained with gradient-based backpropagation.

4. Linear Activation Function

The Linear activation function returns the input directly. Therefore, the output changes proportionally with the input.

$$f(z) = z$$



How it works

- Positive inputs produce positive outputs.

- Negative inputs produce negative outputs.
- The output is not bound to a fixed range.

Advantages

- Simple and computationally inexpensive.
- Easy to differentiate.
- Useful when the output should remain continuous and unbounded.

Limitations

- It does not introduce non-linearity.
- A network made entirely from linear transformations remains linear.
- It is generally unsuitable as the activation function for hidden layers of a deep neural network.

Typical Use

A Linear activation is commonly used at the output layer for regression problems when the target is a continuous, potentially unbounded value.

5. Step vs Linear Activation Function

Feature	Step	Linear	Typical Role
Output	Discrete	Continuous	Depends on task
Shape	Threshold	Straight line	Simple transformation
Differentiability	Not differentiable at threshold	Differentiable	Important for training
Non-linearity	Non-linear	None	Linear is limited
Hidden layers	Generally not used	Generally not used	Non-linear activations preferred
Typical use	Classical perceptron	Regression output	Conceptual foundation

6. Key Takeaways

- An activation function transforms a neuron's weighted sum into its output.
- Non-linear activation functions allow neural networks to learn complex patterns.
- The Step function creates a hard binary threshold and is associated with the classical perceptron.
- The Linear function passes its input directly to the output.
- Linear activation is commonly useful at the output layer for regression, while modern hidden layers generally require non-linear activations.